

Data Flow Diagram (DFD)

Version: 1.0

Project: Crusource CRM

Prepared By: Team Crusource

1. Introduction

The **Crusource CRM** is designed to manage the complete sales lifecycle of an organization — from capturing potential customers as Leads to closing Deals and maintaining customer relationships. The CRM also provides activity tracking, reporting, document management, and AI-powered assistance to improve sales productivity.

A **Data Flow Diagram (DFD)** illustrates how data moves between users, system processes, databases, and external services. It focuses on **how information flows through the system**, rather than how the software is implemented.

2. DFD Symbols

Symbol	Meaning
External Entity	A person or external system interacting with the CRM
Process	A system function that processes data
Data Store	A location where data is stored
Data Flow	Information moving between entities, processes, and data stores

3. External Entities

Entity	Description
Admin	Manages users, permissions, reports, and organization settings.
Sales Manager	Monitors team performance and oversees the sales pipeline.
Sales Representative	Manages leads, contacts, accounts, deals, and daily activities.
Customer	Provides lead information and participates in meetings, calls, and deals.
AI Knowledge Base	Provides AI-powered recommendations, lead insights, and chatbot responses.

4. Data Stores

Data Store	Description
User Database	Users, roles, permissions
CRM Database	Leads, Contacts, Accounts, Deals
Activity Database	Tasks, Meetings, Calls, Notes
Document Storage	Uploaded proposals, contracts, quotations

Data Store	Description
AI Knowledge Base	Embeddings, summaries, chatbot context

5. DFD Level 0 – Context Diagram

DFD Diagram

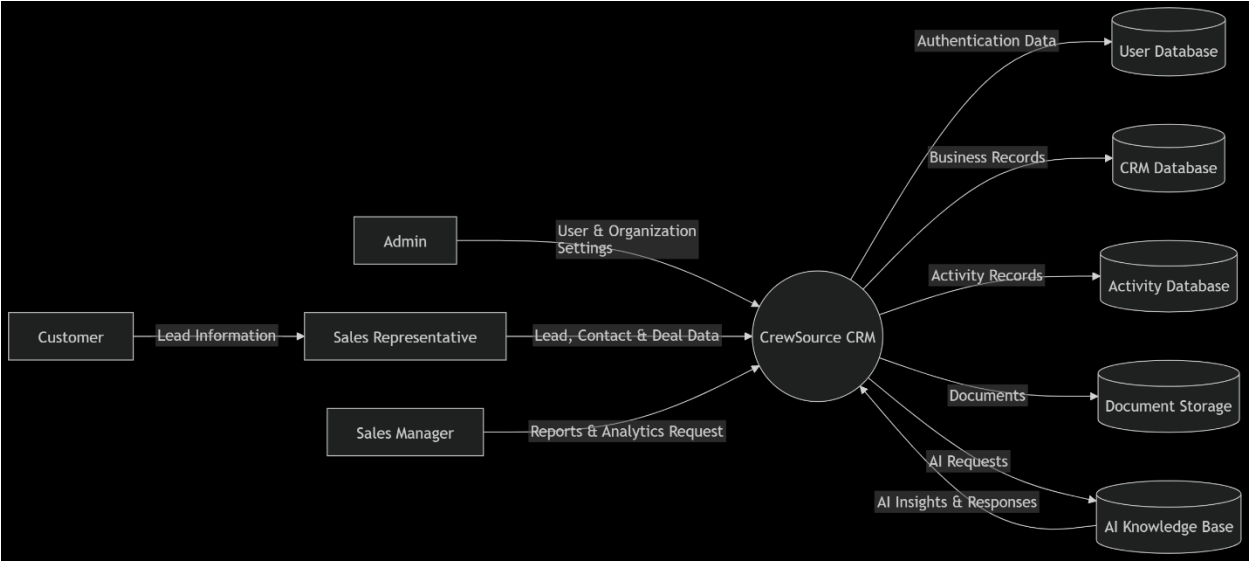


Figure 1: Level 0 Context Diagram

Explanation

The Level 0 Context Diagram represents the Crusource CRM as a single system that interacts with external entities.

- **Admin** manages users, roles, and organizational settings.
- **Sales Representative** performs daily sales operations such as managing leads, contacts, deals, and activities.
- **Sales Manager** monitors team performance and generates reports.
- The **Customer** provides lead information to the **Sales Representative**, who records and manages it within the CRM.
- The CRM processes these interactions, stores information in dedicated databases, and communicates with the AI Knowledge Base to provide intelligent assistance and automation.

This diagram provides a high-level overview of how external entities interact with the CRM without exposing internal business processes.

Data Flow Table

Flow	Source	Destination	Data	Purpose
F1	Customer	Sales Representative	Lead Information	Provide potential customer details
F2	Admin	Crusource CRM	User & Organization Settings	Manage users and settings
F3	Sales Representative	Crusource CRM	Lead, Contact & Deal Data	Perform sales operations
F4	Sales Manager	Crusource CRM	Reports & Analytics Requests	Monitor business performance
F5	Crusource CRM	User Database	Authentication Data	Store user credentials
F6	Crusource CRM	CRM Database	Business Records	Store CRM information
F7	Crusource CRM	Activity Database	Activity Records	Store customer interactions
F8	Crusource CRM	Document Storage	Documents	Store uploaded files
F9	Crusource CRM	AI Knowledge Base	AI Requests	Request AI-powered assistance
F10	AI Knowledge Base	Crusource CRM	AI Insights & Responses	Return AI-generated insights

6. DFD Level 1 – System Overview

DFD Diagram

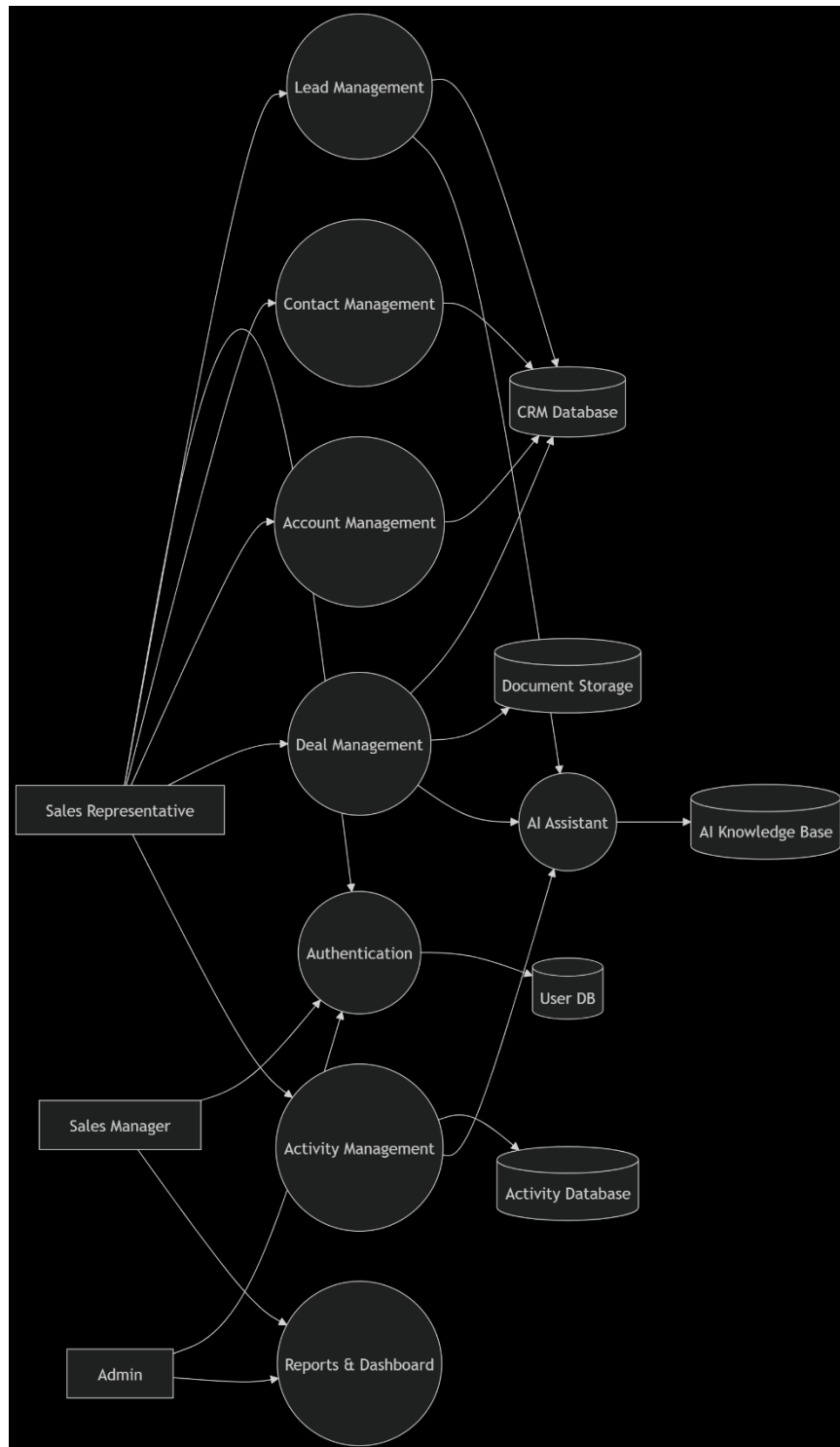


Figure 2: Level 1 System DFD

Explanation

The Level 1 DFD decomposes the CRM into its major functional processes.

Users first authenticate themselves through the Authentication process. Once authenticated, Sales Representatives interact with Lead Management, Contact Management, Account Management, Deal Management, and Activity Management. Sales Managers and Admins primarily interact with the Reporting module to monitor business performance.

Each process exchanges data with the appropriate database, while AI-enabled modules communicate with the AI Knowledge Base to generate intelligent suggestions, summaries, and recommendations.

Data Flow Table

Flow	Source	Destination	Data	Purpose
F1	User	Authentication	Login Credentials	Verify user identity
F2	Authentication	User Database	User Credentials	Validate access
F3	Sales Representative	Lead Management	Lead Information	Manage leads
F4	Sales Representative	Contact Management	Contact Details	Manage contacts
F5	Sales Representative	Account Management	Company Information	Manage accounts
F6	Sales Representative	Deal Management	Deal Details	Manage opportunities
F7	Sales Representative	Activity Management	Calls, Meetings & Tasks	Record customer interactions
F8	Sales Manager / Admin	Reports & Dashboard	Report Requests	Monitor CRM performance
F9	Lead / Contact / Account / Deal Management	CRM Database	Business Data	Store and retrieve CRM records
F10	Lead / Deal / Activity Management	AI Knowledge Base	AI Context	Generate AI responses

7. DFD Level 2 – Detailed CRM Flow

DFD Diagram

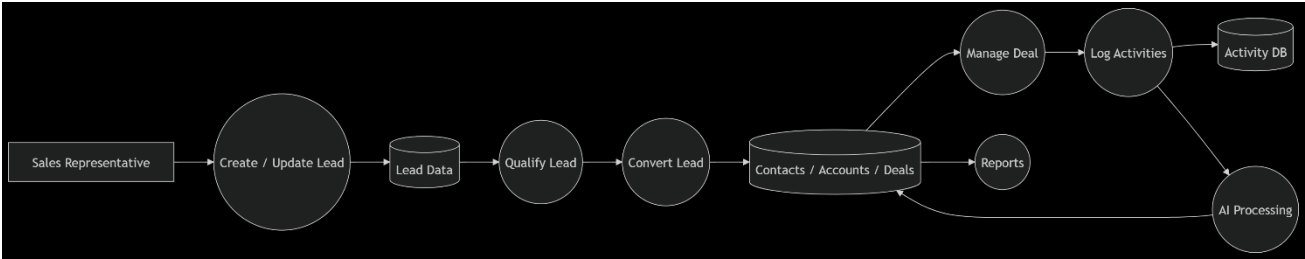


Figure 3: Level 2 Detailed CRM Workflow

Explanation

The Level 2 DFD illustrates the CRM's core business workflow, beginning with lead creation and ending with business reporting.

A Sales Representative creates or updates lead information, which is stored in the Lead Database. Qualified leads are evaluated and converted into Contacts, Accounts, and Deals within the CRM Database. As deals progress through the sales pipeline, all customer interactions are recorded in the Activity Database.

The collected activity data is then processed by AI services to generate intelligent insights, recommendations, and automation. Finally, the processed business data is used to generate reports and analytics for managers and administrators.

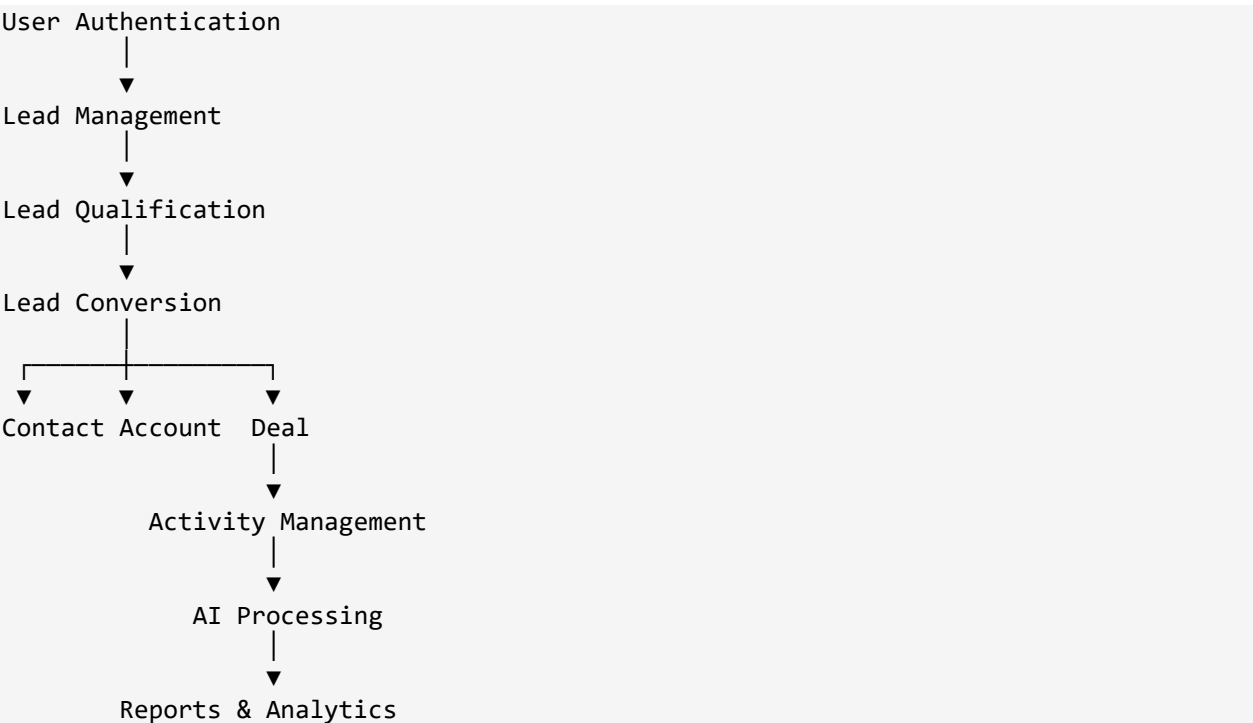
Data Flow Table

Flow	Source	Destination	Data	Purpose
F1	Sales Representative	Lead Management	Lead Information	Create or update leads
F2	Lead Management	Lead Database	Lead Data	Store lead records
F3	Lead Database	Lead Qualification	Lead Details	Evaluate lead quality
F4	Lead Qualification	Lead Conversion	Qualified Lead	Convert lead
F5	Lead Conversion	CRM Database	Contact, Account & Deal Data	Create customer records
F6	CRM Database	Deal Management	Deal Information	Manage sales opportunities
F7	Deal Management	Activity Management	Customer Activities	Track customer interactions
F8	Activity Management	Activity Database	Calls, Meetings & Tasks	Store activities

Flow	Source	Destination	Data	Purpose
F9	Activity Management	AI Processing	Activity Context	Generate AI insights
F10	AI Processing	CRM Database	AI Suggestions	Enhance CRM data
F11	CRM Database	Reports	Business Data	Generate reports and analytics

8. Overall CRM Data Flow

The CRM follows a structured sequence of data movement:



9. Conclusion

The DFD provides a clear representation of how information moves through the Crusource CRM. By progressively refining the system from the Context Diagram (Level 0) to the Detailed Business Process (Level 2), stakeholders can easily understand the interaction between users, system processes, data stores, and AI components. This document serves as a reference for system design, development, and future enhancements.